CASE STUDY | MARKET ENTRY FEASIBILITY

Entering the South Gujarat Ready-Mix Market

How big the opportunity is, what a plant earns, and why the answer is a staged GO

Based on the South Gujarat RMC Market Entry Feasibility Model

Excel workbook, eleven sheets, five districts: Surat, Bharuch, Valsad, Navsari, Tapi

October 2026

In brief

UltraTech is weighing whether to put ready-mix concrete (RMC) plants into South Gujarat. The questions are practical ones: how much concrete the region really buys, who it would be up against, what a plant earns, and what would have to go wrong for the idea to fail. The feasibility model behind this case answers them in a single Excel workbook, and this document walks through how it does that.

The short version. The documented demand pipeline across five districts is worth about ₹26,760 Cr, roughly 44 million m^3. The supply side is crowded but fragmented. A three-plant rollout costing ₹51 Cr returns a base-case NPV of ₹30.8 Cr, an IRR of 40.8% and payback in 2.9 years. The same model turns negative when price, cost and utilisation all move the wrong way together. So the recommendation is a GO, but a staged one: one plant first, and the other two only after the early numbers hold.

Measure	Value
Private projects mapped (GujRERA register)	3,065
Infrastructure projects added	22
Total addressable market (TAM)	₹26,760 Cr | 44.2 million m^3
Serviceable market (SAM)	₹25,943 Cr | 42.7 million m^3
Competitors profiled	66, of which 24 rated High threat
Capital per plant	₹17.0 Cr (₹51 Cr for three)
Base case: NPV, IRR, payback	₹30.8 Cr | 40.8% | 2.9 years
Pessimistic case: NPV, IRR, payback	−₹22.9 Cr | −10.9% | 6.3 years

All figures in ₹ Cr (1 crore = 10 million rupees) unless stated. Volumes are in cubic metres of concrete.

1. The situation

Ready-mix is a short-distance business. Concrete has to be placed within about ninety minutes of mixing, so the model assumes a plant can economically serve sites within roughly 15 to 20 km. Almost every other choice in this case follows from that one physical limit: how much of the market is really reachable, where the plants have to sit, and which competitors count.

The study area is five districts. Surat is the large urban market and holds most of the registered projects. Bharuch brings the Dahej and Ankleshwar industrial belt. Valsad is the coastal strip along NH-48, with the Delhi–Mumbai Expressway packages running through it. Navsari is a developing market with the PM MITRA textile park coming up. Tapi is rural and thin, and a good share of its concrete is still mixed on site.

2. Three questions

The decision is not whether concrete is a good business. It comes down to three narrower questions. First, how much concrete will actually be bought in these districts, and how much of it can one operator reach? Second, who already serves that demand, and how hard will they fight on price? Third, does a plant make money at realistic prices and costs, and what would have to go wrong for it to stop?

The workbook is organised around exactly these three, with a risk register and a one-page summary layered on top.

3. How the workbook is put together

There are eleven sheets, although the cover page is only a placeholder with a date formula. The rest fall into four layers (Figure 1). Inputs feed the market sizing, the market sizing and the price assumptions feed the plant economics, and the results feed a risk register, a dashboard and a summary page.

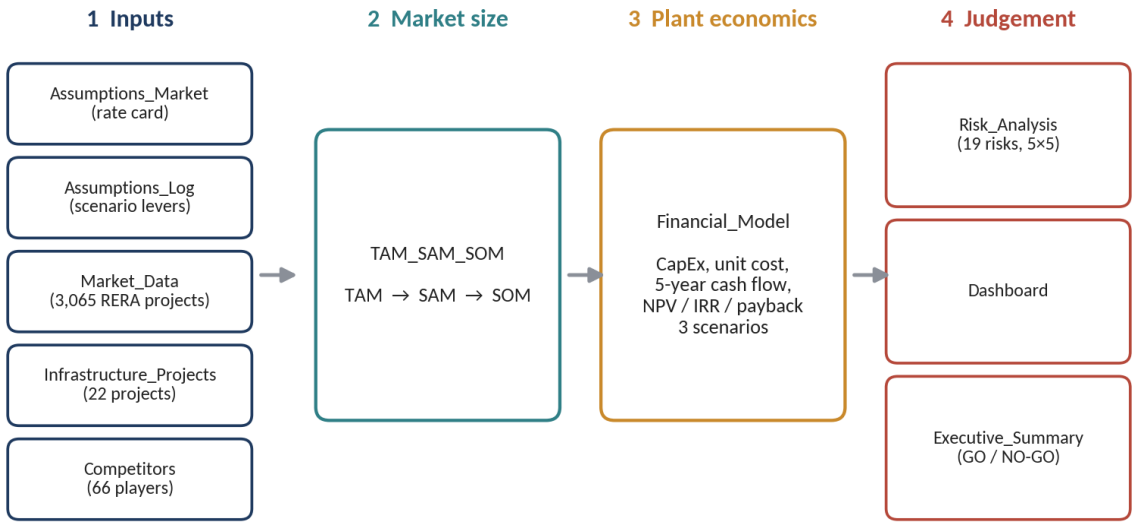


Figure 1. The four layers of the workbook. Arrows show the direction in which numbers flow.

Everything is linked by formula, so changing a price, a utilisation figure or a scenario multiplier carries through to the dashboard. Most of the sentences on the Executive Summary are built with formulas too, which keeps the narrative in step with the numbers. Scenario behaviour is controlled by a handful of named cells on the assumptions log: annual escalation (3%) and multipliers of 1.0x, 2.0x and 0.6x for the base, optimistic and pessimistic market-capture settings.

4. Sizing the market

Starting from the register

The foundation is a list of 3,065 private projects taken from GujRERA, the state RERA register. For each project the model needs a volume of concrete, and the register does not give one, so it is estimated from built-up area. Each project type carries a concrete intensity, in cubic metres of concrete per square metre of built-up area: 0.45 for residential and group housing, 0.50 for commercial, 0.42 for mixed development, and 0.18 for plotted development, which has roads, drains and boundary walls but no buildings. The project is then priced at its district's standard rate.

A worked example from the first row: a villa project in Bharuch with 57,500 sq m of built-up area needs 57,500 × 0.45 = 25,875 m^3 of concrete. At ₹5,800 per m^3 that is about ₹15 Cr of business.

Added up, the private projects come to 37.9 million m^3 and ₹22,960 Cr. The mix is lopsided: Surat accounts for 2,406 of the 3,065 projects (78%) and Tapi for four. A validation column flags projects whose built-up area falls outside the normal range for their category. Of the 3,065, 613 (20%) carry that flag; they stay in the totals but are marked for review.

Adding infrastructure

Twenty-two named public projects sit on a separate sheet: Surat Metro Phase 1, the Bhadbhut barrage, the Delhi–Mumbai Expressway packages in Valsad, the PM MITRA park in Navsari, NH-56 in Tapi and others. Together they add 6.4 million m^3 and ₹3,800 Cr. A formula checks each one against the project register so nothing is counted twice, and screens out anything outside the five districts. Their concrete volumes are benchmark estimates built from published project costs and standard concrete-per-structure ratios. Only one of the 22 is fully verified; 19 are benchmark estimates and two still need validation.

TAM, SAM and SOM

The total addressable market is simply everything above: 3,087 projects, 44.2 million m^3, ₹26,760 Cr. Residential and group housing is the biggest piece, followed by mixed development and the infrastructure block (Figure 2). Industrial and manufacturing sits in the table at zero. The register has no industrial projects, and the model chooses to leave the line empty rather than guess.

The serviceable market applies a reachability rule by district. Surat, Valsad and Bharuch count in full, Navsari counts at 60% because only part of it falls within delivery range, and Tapi is out. That trims ₹817 Cr and leaves a SAM of ₹25,943 Cr, or 97% of TAM. Surat alone is 71% of the reachable demand, which is why it is treated as the first-priority market and why concentration shows up later as a critical risk.

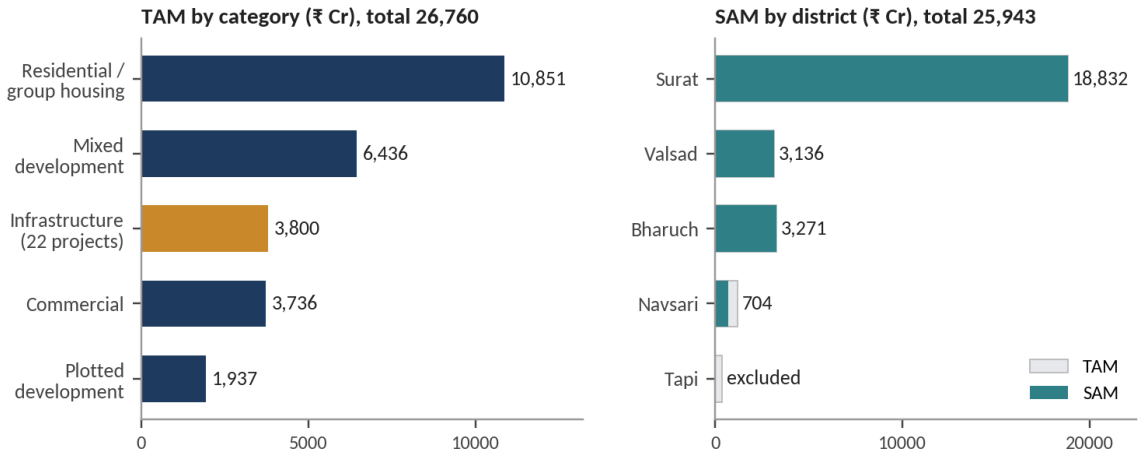


Figure 2. TAM by category and SAM by district, in ₹ Cr. Navsari is counted at 60%; Tapi is excluded.

The serviceable obtainable market is a capture ramp that management has set: 5% of SAM in Year 1, then 10%, 15%, 18% and 20% by Year 5.

	Year 1	Year 2	Year 3	Year 4	Year 5
Capture of SAM	5%	10%	15%	18%	20%
Revenue potential (₹ Cr)	1,297	2,594	3,892	4,670	5,189

These percentages do not drive the financial model, which runs on plant capacity instead. Section 9 comes back to why that matters.

5. The competition

The Competitors sheet profiles 66 operators: 19 in Surat, 14 in Navsari, 13 in Valsad, 11 in Bharuch and 9 in Tapi. Each has an estimated monthly capacity range, a market position, a pricing level, notes on strengths and weaknesses, and a strategic threat rating. Twenty-four are rated High, 26 Medium and 16 Low, and none Very High. Five are captive or project-based plants, the sort an EPC contractor sets up on a big site, and the model keeps those out of open-market share. Only 29 of the 66 entries are marked verified.

Two numbers stand out. The first is price: 39 of the 66 sit in the competitive, cost-effective or low-cost bands, against 8 that price at a premium. The second is distance: each competitor is given its own service radius, roughly 8 to 22 km, and only 11 of the 61 open-market players have the proposed plant location inside that radius. Five of those 11 are High threat. The fight is local. A plant in Surat does not compete with one in Vapi.

The sheet adds a Porter's Five Forces reading and a SWOT. Rivalry, buyer power and the threat of new entrants are all rated High; supplier power and substitutes are Medium. The substitute that matters is captive batching on metro, highway and industrial projects, which takes large volumes out of the open market. The suggested response is selective entry: certified quality and delivery reliability, industrial key accounts around Dahej, Ankleshwar, Hazira and Vapi, mid-premium pricing rather than discounting, and satellite coverage in under-served pockets such as Sarigam, Palej, eastern Tapi and parts of Chikhli and Navsari.

6. What a plant earns

One plant

The base plant is a standard 30 m^3/hour batching plant running ten hours a day for 300 days, which is 90,000 m^3 a year on paper. At the 70% utilisation assumed, it produces 63,000 m^3. That figure, not the market-share percentages in section 4, is what sets revenue in the financial model.

Price is the weighted SAM price, ₹6,069 per m^3. Because Surat dominates the mix, this sits close to Surat's ₹6,200 standard rate and above the standard rate of the other four districts (₹5,800 in Bharuch, ₹5,600 in Valsad, ₹5,500 in Navsari, ₹5,200 in Tapi). Unit cost is built bottom-up at ₹4,600 per m^3, so contribution is ₹1,469 per m^3, or 24.2% of price. Cement alone is ₹2,200, nearly half the stack. Each plant also carries ₹0.6 Cr a year of fixed overhead for site management, sales staff, insurance and depot rent. Price and cost both escalate at 3% a year, which is why the margin percentage stays put and EBITDA grows only through volume. Earlier placeholder margins of 25% and 15% are still visible in the sheet but are flagged as superseded; margin now falls out of price minus cost.

CapEx per plant	₹ Cr
Land / deposit	2.00
Batching plant setup	5.00
Transit mixers	4.00
Concrete pumps	2.00
QA lab / utilities	0.50
Pre-operative cost	0.50
Working capital	3.00
Total	**17.00**

Unit cost	₹ per m^3
Cement	2,200
Aggregates	700
Sand	500
Admixtures	200
Diesel / logistics	500
Employees	250
Maintenance	150
Power / overheads	100
Total	**4,600**

Table: base-case capital and operating cost for one plant.

The three-plant rollout

The plan is one plant in Year 1, a second in Year 2 and a third in Year 3, then hold. Each year's cash flow is that year's EBITDA less the capital for the plant opening the following year, so Year 1 is negative even though the first plant is already earning.

	Plants	Volume (m³)	Revenue	EBITDA	Net cash flow
Year 0	1	–	–	–	−17.00
Year 1	1	63,000	38.24	8.66	−8.34
Year 2	2	126,000	78.77	17.87	0.87
Year 3	3	189,000	121.70	27.66	27.66
Year 4	3	189,000	125.35	28.55	28.55
Year 5	3	189,000	129.11	29.46	29.46
Total			**493.17**	**112.20**	**61.20**

Base case, ₹ Cr except volume. Year 0 is the first plant's capital. The Total row shows cumulative net cash flow over the six periods.

The base case gives an NPV of ₹30.8 Cr at a 12% discount rate, an IRR of 40.8% and a payback of 2.9 years. Five-year ROI, defined as (total EBITDA − total CapEx) ÷ total CapEx, is 120%. The optimistic and pessimistic cases move price, unit cost, utilisation, CapEx and the discount rate together:

	Pessimistic	Base	Optimistic
Price vs base	−5%	₹6,069 per m³	+5%
Unit cost per m³	₹5,060	₹4,600	₹4,370
Capacity utilisation	60%	70%	85%
CapEx per plant	₹18.70 Cr	₹17.00 Cr	₹16.15 Cr
Discount rate	15%	12%	10%
Year 5 revenue	105.1	129.1	164.6
Five-year EBITDA	42.0	112.2	190.4
NPV	−22.9	30.8	93.7
IRR	−10.9%	40.8%	88.6%
Payback	6.3 years	2.9 years	2.1 years
Five-year ROI	−25.2%	120.0%	293.0%

₹ Cr unless stated.

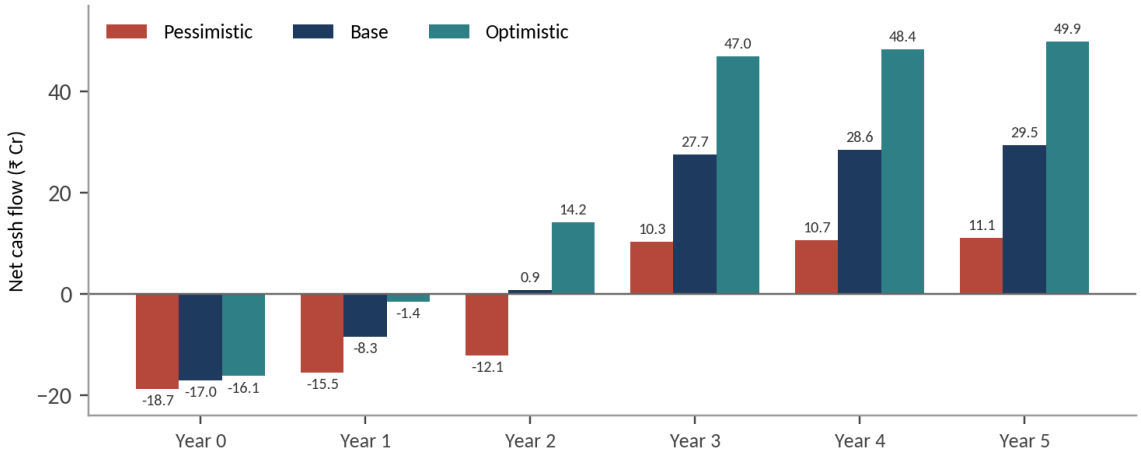


Figure 3. Net cash flow by year under the three scenarios, ₹ Cr. The pessimistic case does not turn positive until Year 3 and takes more than six years to pay back.

7. What can break it

Sensitivity

To see which assumptions carry the case, we reran the base case moving one input at a time across the workbook's own optimistic and pessimistic settings (Figure 4). Before doing that we rebuilt the model outside Excel and checked that it reproduced all three scenario NPVs and IRRs exactly, so the reruns follow the workbook's logic. Unit cost, price and utilisation do most of the damage. CapEx and the discount rate barely matter by comparison.

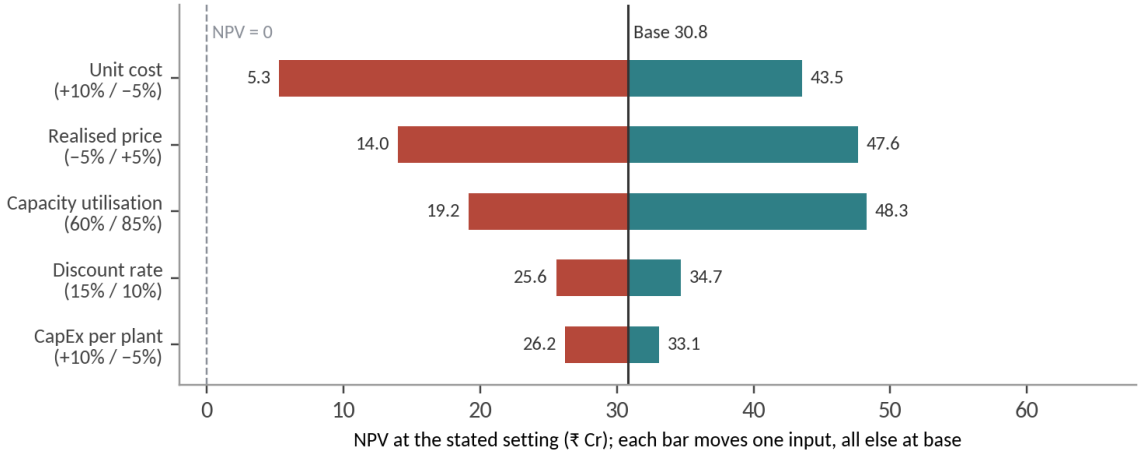


Figure 4. Base-case NPV when one input moves to its pessimistic (red) or optimistic (green) setting. Our own reruns of the workbook logic.

The break-evens are more telling than the bars. With everything else at base, NPV reaches zero if the realised price falls about 9% to roughly ₹5,514 per m^3, if unit cost rises about 12% to roughly ₹5,156 per m^3, or if utilisation drops to about 44%. CapEx per plant would have to reach about ₹28 Cr, two-thirds above the estimate. Two modest moves together are enough: a 5% lower price combined with a 10% higher unit cost takes NPV to about −₹11.5 Cr with every other input unchanged. The cushion is thin where it counts, because the whole case rests on a ₹1,469 spread on a ₹6,069 sale.

Risks

The risk register holds 19 entries, each scored for probability and impact on a 5 × 5 grid (Figure 5). Three are Critical, six High, ten Medium and none Low. The three Critical ones are plant location and service-radius selection, dependence on Surat demand, and whether market capture ramps up as planned.

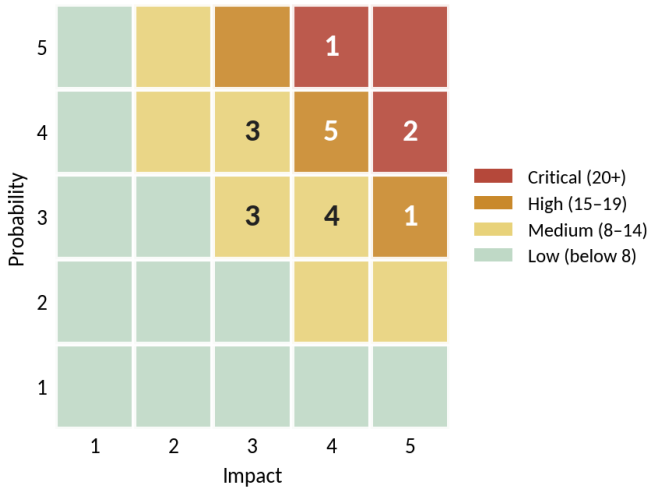


Figure 5. Count of risks by probability and impact. Severity is probability × impact; the average across the 19 risks is 14 out of 25.

Most of the mitigations in the register are about sequencing rather than spending. Fix the site only after mapping the SAM districts, competitor locations and delivery lanes. Sign anchor builders in Surat before committing capacity, then widen to Valsad and Bharuch. Clear permits before buying land. Lock cement, aggregate and sand terms before volumes ramp. Tie each new plant to signed onboarding milestones.

8. The recommendation

The decision cell on the Executive Summary reads GO, and the text around it is specific that this means proceeding on a milestone-gated basis. Year-1 capital goes to one plant, ₹17 Cr. After that, capital is released against evidence.

Step	What has to hold before releasing the money
Plant 1 (Year 0)	Site confirmed after mapping districts, competitor locations and delivery lanes; permits cleared; anchor builders signed in Surat.
Plant 2 (Year 1 to 2)	Plant 1 running near the 70% utilisation assumption; realised price close to ₹6,069 per m^3; booked demand tracking the base case.
Plant 3 (Year 2 to 3)	The same tests again, with orders starting to come from Valsad and Bharuch so that dependence on Surat eases.
Hold trigger	If price, utilisation or cost drift toward the pessimistic figures, pause expansion and retest NPV before spending further.

Table: stage gates as set out in the risk mitigations and strategic recommendation.

9. Where the analysis is thin

The workbook is candid about a lot of its own limits. It tags every input as Validated, Benchmark Estimate or Needs Validation, and its dashboard shows how lopsided the data sample is. A few further points are worth raising before anyone relies on the numbers.

TAM is a pipeline, not a yearly market. The register runs from projects started in 2005 to ones finishing in 2036, and 54% of the demand sits in projects already marked completed. The ₹26,760 Cr is therefore the lifetime value of everything documented, not a year of buying. A rough annualisation, which we did outside the workbook by dividing each ongoing private project's demand by its build duration, gives something like 3.5 million m^3 a year across the five districts, against the 44 million in the TAM. The financial model is unaffected because it runs on plant capacity, but the SOM percentages inherit the problem.

The SOM and the plant plan describe different things. The SOM puts Year 1 at 5% of SAM, ₹1,297 Cr. The plant plan earns ₹38 Cr in Year 1, about 0.15% of SAM volume, and roughly 0.44% by Year 3 when all three plants run. The assumptions log itself records that an earlier version had exactly this scale mismatch and was corrected by switching to capacity-based volumes. The market capture and annual volume rows at the top of the financial sheet, which still show about 2.1 million m^3, are no longer used in the forecast. Against the annualised demand above, three plants producing 189,000 m^3 would be around 5% of the market, which is a more demanding and more honest test of whether the customers exist.

SAM is drawn by district, but delivery works by radius. Reachability is Yes, Partial or No by district, which leaves SAM at 97% of TAM. Yet the assumptions log sets the economic delivery radius at 15 to 20 km, and the competitor sheet shows 50 of the 61 open-market competitors too far from the proposed plant to reach it, with distances running up to about 103 km. One plant cannot serve three districts. A SAM drawn as a radius around real plant sites would be much smaller, and it would make the plant-location risk easier to size.

Prices are desk estimates. Every district rate is tagged as an internal desk estimate with no vendor rate card on file, and the sheet itself recommends checking recent quotations before a board submission. With 39 competitors in the lower price bands, and a break-even price only about 9% below the model's, this is the first thing to test. A related question is whether the ₹2,200 cement line is a market price or an internal transfer price, since that moves unit cost directly.

Competitor shares are shares of the wrong total. Each competitor's share is its capacity divided by the capacity of all 66 listed players across all five districts, so the shares add to 100% overall. The zone totals then simply show each district's slice (Surat 33%, Bharuch 28%, Navsari 22%, Tapi 10%, Valsad 7%), and the implied "unorganised / other" residual per district, such as 67% for Surat and 93% for Valsad, is just the remainder of that arithmetic. It is not an estimate of the informal market. Share should be worked out inside each district against that district's demand.

There is no ramp-up. Year 1 assumes 70% utilisation from the first day. If each new plant instead ran at 50%, then 65%, then 70% over its first three years, base NPV falls to about ₹22.8 Cr and IRR to about 32%.

Cash flow is EBITDA less CapEx. There is no tax, depreciation, interest, working-capital release or terminal value, so NPV and IRR are pre-tax project figures and best read as directional. Two of those omissions, terminal value and working-capital release, lean conservative. Tax and the missing ramp-up lean the other way.

Data coverage is uneven. Surat is 78% of registered projects and Tapi has four. Twenty per cent of projects are flagged for built-up-area review, 19 of 22 infrastructure volumes are benchmark estimates, and 37 of the 66 competitors are unverified. District-level results outside Surat should be read as directional.

10. What stays with us

Geography is the market. In ready-mix, a district total means little until it is tied to a plant location and a delivery radius.

Capacity, not share, set the revenue. The numbers became believable once the model was rebuilt around what a plant can physically produce, and they came out far smaller than the top-down view suggested.

Two inputs decide the case. Price and unit cost, with utilisation close behind. The whole decision rests on a ₹1,469 spread, and cement is nearly half of the cost.

Label what you know. Tagging inputs as Verified, Benchmark Estimate or Needs Validation costs little and makes a reviewer's job much easier.

Stage the money. A GO that releases capital plant by plant is far cheaper to be wrong about than one that commits ₹51 Cr on day one.

Appendix A. Sheet-by-sheet guide

Sheet	What it does	Technique worth learning
Cover_Page	Title page; holds only a date formula.	TODAY()
Assumptions_Market	Concrete intensity by project type; district price rate card with min, max and standard price.	Excel tables used as lookup sources
Market_Data	3,065 RERA projects: built-up area to demand to value; status and validation flags; district summary.	XLOOKUP, SUMIF, COUNTIFS on a structured table
Infrastructure_Projects	22 public projects; duplicate and geography check; demand tiers.	INDEX / MATCH, nested IF, SUMIFS
TAM_SAM_SOM	Totals by category, district reachability, capture ramp.	SUMIFS across two tables; reachability lookup
Competitors	66 players; capacity read from text ranges; threat and pricing scores; Porter and SWOT.	LEFT / MID / FIND text parsing; scoring flags
Financial_Model	CapEx, unit cost, plant capacity, 5-year forecast, three scenarios, NPV, IRR, payback.	NPV, IRR, nested-IF payback; scenario columns
Risk_Analysis	19 risks scored probability × impact; heat map and count matrix.	COUNTIFS grid; severity bands
Assumptions_Log	Every assumption with source, status and sensitivity; named scenario cells.	Named ranges (escalation, scenario multipliers)
Dashboard	KPIs, funnel and charts, data-quality table, recommendation banner, risk heat map.	Linked cells and charts
Executive_Summary	Narrative built from live numbers; GO / NO-GO dropdown.	TEXT() string building; data validation

Appendix B. Notes on method

Market, competitor, cost and scenario figures are taken directly from the workbook as it stands. The sensitivities in section 7, the annualised-demand estimate and the roughly 5% comparison in section 9, and the ramp-up case are our own calculations outside the workbook. Before running them we rebuilt the financial logic and confirmed that it reproduced the base, optimistic and pessimistic NPVs and IRRs exactly. The annualised demand figure is approximate: it covers ongoing private RERA projects only and divides each project's estimated demand by its planned build duration. Project status in the workbook is calculated against today's date, so the completed and ongoing counts will shift over time. The competitor sheet's own "Within Radius" flag counts 16 players; five of those are captive plants with no radius, which the formula passes by default, so we use 11 for open-market competitors.